Study Smarter, Not Harder: Student Edition speaker guide

Deck: `presentation/Study_Smarter_Not_Harder_Student_Edition.html` (11 main slides, then backups A1 to A5). About 10 minutes.

Keys: right arrow or Space = next. Left arrow = back. **1 / 2** = count a hand on slides 1 and 8. **Z** = include or exclude the zero-grade students on slides 5 and 6. **N** = stats details (only if asked). **R** = recap. **A** = backups.

The presentation in one sentence

We look at 649 students' real grades to answer one question: does studying pay off, and does it still pay off once we compare like with like? The short answer is yes, but history matters more, and fifteen students with perfect attendance and a zero grade nearly fooled us.

Who talks when

Speaker	Slides	About
Kartik	1 (the opening vote), 2 (the data), 3 (Act 1: effort shows up)	2:15
Jolae	4 (Act 2: compare like with like)	1:30
Devsen	5 (the Perfect Attendance Paradox), 6 (the maths check)	2:10
Harsh	7 (the line-up), 8 ("Your turn", Student A vs B)	1:55
Nikhil	9 (did we break any rules?), 10 (the verdict), 11 (what we can't claim)	2:00

Total is about 9:50. Cap each vote at 30 seconds. If the room is slow, skip the hands and just give the answer.

The 11-slide map

1. **The opening vote:** two made-up students, who gets the higher grade?
2. **The data:** 649 students, three grades, one self-reported study bracket.
3. **Effort shows up:** the study-time gap is 2.38 points.
4. **Compare like with like:** the effect shrinks as we add controls.
5. **The zero-grade paradox:** the fifteen zeros had zero absences.
6. **Does it happen again?** the same pattern in the maths course.
7. **The line-up:** what else predicts a final grade.
8. **Your turn:** Student A or Student B?
9. **Did we break any rules?** the checks behind the numbers.
10. **The verdict:** effort shows up, history dominates, check your zeros.
11. **What we can't claim:** limits, then questions.

Kartik · slides 1 to 3

Slide 1: The opening vote

Talking anchors

- Two made-up students, the Grinder and the Regular.
- Grades are out of twenty, and the vote is about who ends up higher.
- Don't reveal anything yet. Just get people thinking.

Natural delivery "Two students, both made up for this. The Grinder studies ten hours a week but misses twenty classes. The Regular studies two hours and never misses one. Grades are out of twenty. Hands up: who ends up with the higher final grade?"

Count hands with 1 and 2, but keep it to about 30 seconds.

"Hold on to that. By the end you'll see why perfect attendance can hide a zero. The answer comes from real records of 649 students."

Transition: "Here's what those records look like."

Memorize this

> Two made-up students, one vote, and a promise that the real data will settle it.

Slide 2: The data

Talking anchors

- 649 students, a Portuguese-language course, two schools, from Cortez in 2008.
- Three grades each: G1, G2 and G3. G3 is the final, and it's what we're trying to explain.
- Study time is a tick-box, not hours. We watched students; we didn't assign them to study.
- Fifteen students got exactly zero. We'll come back to them.

Natural delivery "We have 649 students in a Portuguese course, at two schools. Each has three grades out of twenty. The last one, G3, is what we're trying to explain. The average is about 11.9, and the usual spread is about 3.2 points."

"Two warnings before we start. Study time was a box the student ticked: under 2 hours, 2 to 5, 5 to 10, or over 10. It wasn't a stopwatch. And nobody was assigned to study. So everything today is a pattern, not proof."

Point at the orange bar. "Fifteen students scored exactly zero. Nobody scored 2, 3 or 4. Remember that."

Transition: "So, does studying pay?"

If challenged "Does this mean studying causes better grades?" "Not on its own. It's an association. Nobody was assigned to study."

Memorize this

> Real data, a self-reported study box, and a pattern we watched rather than caused.

Slide 3: Does studying pay?

Talking anchors

- Average grade climbs from about 11 for the lightest studiers to about 13 for the 5 to 10 hour group.
- The gap is 2.38 points, with a 95% interval of 1.40 to 3.37.
- The chart's axis starts at 9, so the bars look taller than the gap really is. Say so out loud.
- ANOVA says the difference is not luck: p is about 6 in 10 billion. Study time explains about 7% of the variation.
- Past ten hours there's no extra gain. Only 35 students are up there, so go easy on that end.

Natural delivery "Do students who study more get better grades? Yes. The average goes from about 11 for the lightest studiers to about 13 for the 5 to 10 hour group. Quick note: the axis starts at 9, so the bars look taller than the gap really is."

Click for the gap. "The gap is 2.38 points out of 20. The usual spread is 3.2, so it's about three-quarters of one spread. The 95% interval runs from 1.40 to 3.37."

"Is that luck? Very unlikely. ANOVA gives p of about 6 in 10 billion. Study time explains about 7% of the variation in grades. Small, but real."

Click for the plateau. "Past ten hours, no extra gain. Minus 0.17. Only 35 students are up there, so don't read too much into that end."

Transition: "But who are the students who study more? Jolae makes it a fair comparison."

If challenged "Why not just say 'the effect is 2.4 points'?" "Because the interval matters. It runs from about 1.4 to 3.4. We're not sure of the exact number, only the range."

Memorize this

> Students who study more score about 2.4 points higher, and that gap is not chance.

Jolae · slide 4

Slide 4: Compare like with like

Talking anchors

- Raw comparisons mix studying with who studies.
- Step by step, we add controls: M1 is study time alone, M2 adds academic and demographic factors, M3 adds family and social factors, and M4 adds earlier grades.
- The coefficient goes from +0.97 to +0.45 to +0.42 to +0.05.
- M3 is our main model. M4 undersells effort because the earlier grades were partly earned by studying.

Natural delivery "Is it the studying, or the kind of student who studies? Think of judging fertiliser. You compare plants that got the same sun and water first."

"Comparing everyone with everyone, each step up a study bracket goes with about +0.97 points."

Click for M2. "Now match students on past failures, absences, sex, age, school, address and whether they want higher education. The effect more than halves, to about +0.45. Most of the shrinking happens right here."

Click for M3. "Now also match on family and social life. It barely moves: +0.42, p = 0.001. This is our main model. A small, real effect."

Click for M4. "Add earlier grades, G1 and G2. The effect drops to +0.05, and R-squared jumps from 0.34 to 0.86."

Pause. "Careful. Those earlier grades were partly earned by studying. So this last step undersells effort. It doesn't say effort is useless. Past grades matter far more, but effort still shows up."

Transition: "One thing wouldn't behave in any of these models: missing class. Over to Devsen."

If challenged "Why does the effect vanish in M4?" "Because G1 and G2 are partly caused by studying. Controlling for them hides some of the effect. ANCOVA with G1 gives p = 0.097, same story."

Memorize this

> Compare like with like, and the effect shrinks from about one point to about four-tenths. History carries most of the weight.

Devsen · slides 5 and 6

Slide 5: The Perfect Attendance Paradox

This is the peak of the talk. Slow down here.

Talking anchors

- With everyone included, absences look harmless: minus 0.15, not significant.
- All fifteen zero-grade students had zero recorded absences.
- Among everyone else, fewer than four in ten had zero absences. Fifteen out of fifteen by chance is about four in ten million.
- Their grades were already sliding: about 7, then about 4, then zero. Seven already had a zero in period two.
- Remove them and ten absences cost 1.4 points, not 0.6. The model coefficient goes from minus 0.15 to minus 0.45, now strongly significant.
- Our reading is that they stopped coming and got marked off the register. The data never say why. Call them apparent dropouts, and say that's our interpretation.

Natural delivery "You'd expect missing class to hurt. With everyone in, it barely seems to. Ten missed classes cost only about 0.6 points, and in our main model the effect is minus 0.15."

Click, or press Z, to highlight the zeros. "Look at the orange dots in the bottom left. All fifteen students who scored zero had zero recorded absences. Perfect attendance, zero grade."

"Among everyone else, fewer than four in ten have zero absences. The chance that fifteen out of fifteen would have none at random is about four in ten million."

"Their grades were already sliding. About 7 in the first period, about 4 in the second, then zero. Seven of them already had a zero in period two. Nobody else did."

Click, or press Z again, to remove them. "Take them out, and ten missed classes cost 1.4 points, more than double. In the full model the absence effect goes from minus 0.15 to minus 0.45, and it's now strongly significant."

"Here's our reading. They stopped coming, and a register can only mark you absent while you're still on it. We call them apparent dropouts. That's our interpretation, though. The data never say why."

Press Z once more so the room sees the line flip back.

Transition: "Could this be a quirk of one file? Let's check."

If challenged "Isn't dropping the zeros cherry-picking?" "No, and we show both. A zero-grade dummy keeps all 649 students and gives minus 0.47. Keeping only students with a period-two grade gives minus 0.28. Every version is negative."

Memorize this

> Fifteen students with perfect attendance and a zero grade were hiding the cost of missing class.

Slide 6: Does it happen again?

Talking anchors

- Same pattern in the maths course, which has 395 students.
- All 38 maths zero-grade students had zero recorded absences.
- With them in, skipping class looks helpful: plus 1.7 points for ten absences. Without them it flips to minus 1.9.
- The school pattern did not repeat, so we make no claim about schools.

Natural delivery "Could this just be one file? We checked the maths course. Thirty-eight students scored zero in maths, and all thirty-eight had zero recorded absences."

Click, or press Z. "With them in, skipping class even looks helpful: plus 1.7 points for ten missed classes. Take them out and it flips to minus 1.9."

"One thing didn't repeat. In Portuguese, 14 of the 15 zeros were at one school. In maths they're spread about in line with school size. So we make no claim about schools."

Transition: "So what does predict a final grade? Harsh."

Memorize this

> The same flip shows up in a second course, so this isn't a fluke of one file. The school story doesn't hold.

Harsh · slides 7 and 8

Slide 7: The line-up

Talking anchors

- Wanting higher education is the biggest single difference: about 3.5 points raw, about 1.8 after comparing similar students.
- Each past failure costs about 1.4 points.
- Going from under 2 hours to 5 to 10 hours is worth about 0.8.
- Family support shows nothing clear.
- Extra school support looks negative, minus 1.3. That's selection, not harm.

Natural delivery "So what does predict a final grade? The biggest single difference is ambition. Students who want higher education score about 3.5 points higher. That's more than one usual spread. Compare similar students, and it's still about 1.8."

"Each past failure costs about 1.4 points. Moving from under 2 hours to 5 to 10 hours is worth about 0.8. Family support shows nothing clear."

Point at the school-support bar. "Extra school support looks negative, minus 1.3. That isn't help hurting. Think of hospitals: the sickest patients get the most treatment. Support goes to students who are already struggling."

"Cause and effect can run both ways here too. Struggling students may study more, or give up. We can name that, but we can't untangle it with this data."

Transition: "Let's test it on two real-looking students. Your turn."

If challenged "Is the school-support effect real?" "It's a selection effect, not a measure of support. The model can't tell us what support would have done."

Memorize this

> Ambition and past failures move grades more than an extra study bracket does.

Slide 8: Your turn

Talking anchors

- Student A studies over ten hours, has 12 absences, one past failure, no extra school support.
- Student B studies 2 to 5 hours, has 2 absences, no failures, and gets extra school support.
- The model says A, 12.6 to 12.2. That's almost a tie.
- A real student with either profile could land anywhere from about 7 to 18. The model predicts the average, not the person.
- A's lead comes only from B's school support. Treat them the same and B leads by 0.93.

Natural delivery "Your turn. Student A studies over ten hours, has 12 absences, one past failure, and no extra school support. Student B studies 2 to 5 hours, has 2 absences, no failures, and gets extra school support."

"Hands up. Who does the model think will score higher?"

Count hands with 1 and 2, under 30 seconds.

Click to reveal. "A, barely. 12.6 against 12.2. Almost a tie."

Click for the spread. "Look at the thin lines. A real student with either profile could land anywhere from about 7 to 18. A model predicts the average student, not any one person."

Click for the twist. "A is ahead only because B gets extra support, and support marks a struggling student. Treat them the same on support, and B leads by 0.93."

Transition: "Before we trust any of this, Nikhil checks whether we broke the rules."

If challenged "What's the prediction range for one student?" "About 7 to 18. That's wide, which is the honest answer for one person."

Memorize this

> A profile predicts an average, not a person. Two students with the same profile can land far apart.

Nikhil · slides 9 to 11

Slide 9: Did we break any rules?

Talking anchors

- Seven checks, each written as a plain question.
- Are the predictors tangled together? No. The biggest VIF is 1.99.
- Does one student drive the results? No. The biggest Cook's distance is 0.037.
- Do the errors follow a bell curve? Yes, once the zeros are set aside.
- The scatter isn't perfectly even, so we use robust error bars (HC3) everywhere. With the zeros in, the test fails at p of 3.5 in 10 billion. Without them, p is 0.24.
- A straight line isn't a perfect fit, probably because grades stop at 0 and 20. We report that openly.

Natural delivery "Every regression comes with rules. We checked seven, each as a plain question."

"Are the predictors tangled together? No. The biggest VIF is 1.99. Does one student drive the results? No. The biggest Cook's distance is 0.037. Do the errors follow a bell curve? Yes, once the zeros are set aside."

"Two things to watch. The scatter around the line isn't perfectly even, so we used robust error bars, HC3, on every result. With the zeros in, that test fails badly. Without them it passes. And a straight line isn't a perfect fit, probably because grades are capped at 0 and 20. We say that openly."

"Row order doesn't matter here. This is one snapshot of students, not a time series."

Transition: "So what's the verdict?"

If challenged "Is the RESET failure fatal?" "It means a straight line isn't exact. Grades are capped at 0 and 20, so that's expected. The main conclusions hold with and without the zeros."

Memorize this

> The model mostly passes the checks. The one red flag is reported, not hidden.

Slide 10: The verdict

Talking anchors

- Effort shows up. History dominates. Check your zeros.
- Effort: +2.38 raw, and about +0.42 per step after controls.
- History: R-squared goes from 0.34 to 0.86 when earlier grades join.
- Zeros: all fifteen had zero absences, and without them the absence effect is minus 0.45.
- For you: your last grade is the best guide to your next, so change course early.

Natural delivery "Here's the verdict in three lines. Effort shows up. History dominates. Check your zeros."

"On effort: plus 2.38 raw, and about plus 0.42 per step after controls. On history: R-squared jumps from 0.34 to 0.86 once earlier grades join. On zeros: all fifteen zero grades had zero absences, and without them the absence effect is minus 0.45."

Click for the vote tally. "Back to your vote. The data credit both studying and showing up. And the perfect attendance record is exactly where the zero grades were hiding."

"For you, the practical bit: your last grade is the best guide to your next. So if you're slipping, change course early."

Transition: "Before questions, let's be clear about what we can't claim."

Memorize this

> Effort shows up, history dominates, and before you believe a regression, check its zeros.

Slide 11: What we can't claim

Talking anchors

- These are patterns, not proof.
- Students chose how much to study.
- Earlier grades undersell effort.
- "Apparent dropouts" is our reading. Every key result is shown with and without the zeros.
- Study time is four tick-boxes, from two schools, in one year.

Natural delivery "What we can't claim. These are patterns, not proof. Students chose how much to study. Earlier grades undersell effort. The dropout story is our reading. Study time is four tick-boxes. And it's two schools in one year."

"Thank you. Questions?"

Stop. Press A if you need the backups.

Memorize this

> Patterns, not proof. Say that before anyone else does.

Defense Q&A

Memorise the logic, not the wording.

1. **Does 2.4 points mean studying causes better grades?** No. Students weren't assigned to study, and study time is self-reported on four levels. It's an association.
2. **Why Tukey after ANOVA?** ANOVA says some groups differ. Tukey says which pairs, and it keeps the false-positive rate at 5% across six comparisons. Levene says the spreads are similar, and Kruskal-Wallis agrees without assuming a bell curve.
3. **Is the study effect real after controls?** Small but yes. In M3 it's +0.42, p = 0.001. Without the zeros it's +0.31, p = 0.009. With three dummies instead of a straight line, it's borderline without the zeros (p = 0.063), so we call the evidence modest. Backup A3 has the detail.
4. **Why does it vanish in M4?** Earlier grades are partly caused by studying, so controlling for them is a "bad control". It hides effort. ANCOVA with G1 gives p = 0.097.
5. **Isn't dropping the zeros cherry-picking?** We show both. The zero-grade dummy keeps all 649 and gives minus 0.47. Keeping only students with a period-two grade gives minus 0.28. Every version is negative. Backup A4 has the detail. The claim is limited to students who completed the course.
6. **Is maths an independent replication?** No. Many students appear in both files, so it's a consistency check, not proof.
7. **Why is school support negative?** Selection. Support goes to students who are already struggling. The coefficient is minus 1.33 in M3.
8. **Why a range of 7 to 18 for one student?** That's a prediction interval for one person. It's wide because people vary a lot around any average.
9. **Is the RESET failure fatal?** No. It means the straight line isn't exact, which is expected for grades capped at 0 and 20. The conclusions hold with and without the zeros.
10. **Why HC3?** It's a conservative robust correction for small samples. It changes significance for at most one of the seven key terms here.
11. **Can the model replace judgement?** These are patterns, not predictions you can bank on. Treat the findings as a reason to check, not a verdict on any one student.

The 2-hour prep method

- **0 to 20 min:** read the one-sentence summary, the slide map and the "Memorize this" lines. Don't memorize paragraphs.
- **20 to 50 min:** for each of your slides, say the "Memorize this" line from memory, then explain the talking anchors without looking.
- **50 to 75 min:** do one full run, standing, with the slides on screen. No reading. Focus on smooth handoffs.
- **75 to 100 min:** defense drill. One person plays the professor and asks the Q&A questions. Answer in 20 to 40 seconds. If you get stuck, go back to the principle, not invented details.
- **100 to 120 min:** full run at speed. Fix only obvious gaps. Don't rewrite everything at the last minute.

Confidence rules

- Look at the slide, look at the audience, then explain the idea.
- If you blank, return to the "Memorize this" line and carry on.
- Say "association," "pattern," or "probability," not "proves."
- If you're asked for a number you don't have, say the results file has it. Don't guess.

Notes for the team

- The Student Edition has 11 main slides (the Story Deck has 14). The embedded notes in the HTML assume four speakers, so this guide re-cuts them for five.
- The opening is a hands-up vote on The Grinder vs The Regular, not the Story Deck's numerical guess. Slide 3's 2.38 is not tied back to a guess.
- The Story Deck's "who studies" slide (77% girls, a third as many failures) is not a main slide here. Sex appears only in backup A2.
- The Story Deck's "fading zeros" slide and maths cross-check (22 of 31, 26 passed) are not in this deck.
- The absence slides from the Story Deck are merged into slide 5, with the Z toggle.
- The verdict (slide 10) comes before the limits (slide 11). Both go to Nikhil.
- Probability of 15 of 15: this deck shows 4e-7, so say "about four in ten million". The Story Deck quotes 3.2e-7 ("3 in 10 million"). Pick one and stick with it.
- The maths figures on slide 6 are the simple-line numbers (+1.7 to -1.9). The Story Deck quotes the M3 numbers (+0.98 to -0.55). Both are in the results files, so know which one you're saying.
- The 0.83 "fair gap" for the same two students is on slide 7 here, but it isn't repeated in the verdict.
- Slide 10 reads out the opening-vote tally, so make sure someone counted hands on slide 1.